Class 12 Mathematics

Chapter 3: Matrices

Practice Set B: step-wise solutions

20 questions **49** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ PYQ-style — 1 mark

If A is a matrix of order $m \times n$ and B is a matrix such that AB' and $B'A$ are both defined, then the order of B is:

(A) $m \times m$ **(B)** $n \times n$

(C) $n \times m$ **(D)** $m \times n$

Formula / idea: $P_{a\times b}\,Q_{c\times d}$ is defined only if $b = c$.

1 Let B be of order $u \times v$, so B' is $v \times u$.

2 $A_{m\times n}B'_{v\times u}$ is defined.
$n = v$

3 $B'_{v\times u}A_{m\times n}$ is defined.
$u = m$

4 So B has order $u \times v = m \times n$. Option (D). ✓ 1

Answer **(D)** $m \times n$

Watch out: Finding the order of B' ($n \times m$) and forgetting to transpose back to B.

2 MCQ PYQ-style — 1 mark

If $\begin{bmatrix} x+y & 2 \\ 5 & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$, then the value of $\left(\frac{24}{x} + \frac{24}{y}\right)$ is:

(A) 3 **(B)** 4

(C) 6 **(D)** 18

Formula / idea: Equal matrices have equal corresponding elements.

1 Equate corresponding elements.
$x + y = 6, \qquad xy = 8$

2 Take 24 common and use a common denominator. ✓ 1

$$\frac{24}{x}+\frac{24}{y}=\frac{24(x+y)}{xy}=\frac{24\times 6}{8}=18$$

3 Hence option (D).

Answer **(D)** 18

Watch out: Writing $\frac{24}{x}+\frac{24}{y}=\frac{24}{x+y}=4$; fractions do not add like that.

3 MCQ NCERT Ex 3.3 Q11 1 mark

If A and B are symmetric matrices of the same order, then $AB-BA$ is a:

(A) **skew-symmetric matrix** **(B)** symmetric matrix

(C) zero matrix **(D)** identity matrix

Formula / idea: $(AB)'=B'A'$; symmetric means $A'=A$.

1 Take the transpose of $AB-BA$.

$$(AB-BA)'=(AB)'-(BA)'=B'A'-A'B'$$

2 Use $A'=A$ and $B'=B$. ✓ 1

$$(AB-BA)'=BA-AB=-(AB-BA)$$

3 So $AB-BA$ is skew-symmetric. Option (A).

Answer **(A) skew-symmetric matrix**

Watch out: Writing $(AB)'=A'B'$ and concluding the matrix is symmetric.

4 MCQ Board-pattern 1 mark

Let $A=[a_{ij}]$ be a 3×2 matrix whose elements are given by $a_{ij}=(i-3j)^2$. The element in the 2nd row and 3rd column of A' is:

(A) 0 **(B)** 9

(C) 16 **(D)** 49

Formula / idea: The (i,j) element of A' is the (j,i) element of A.

1 A is 3×2, so A' is 2×3 and has a $(2,3)$ element.

2 Swap the indices.

$$(A')_{23}=a_{32}$$

3 Put $i=3$, $j=2$ in the rule. ✓ 1

$$a_{32}=(3-6)^2=9$$

4 Hence option (B).

Answer **(B)** 9

Watch out: Putting $i = 2, j = 3$ in the rule to get 49; a_{23} does not even exist in A.

5 **MCQ** NCERT Exemplar 1 mark

If A is a square matrix such that $A^2 = I$, then $(A - I)^3 + (A + I)^3 - 7A$ is equal to:

(A) A **(B)** $I - A$

(C) $I + A$ **(D)** $3A$

Formula / idea: A and I commute, so the binomial expansion holds; $A^3 = A \cdot A^2 = A$.

1 Expand both cubes.
$(A - I)^3 = A^3 - 3A^2 + 3A - I$
$(A + I)^3 = A^3 + 3A^2 + 3A + I$

2 Add; the A^2 and I terms cancel.
$(A - I)^3 + (A + I)^3 = 2A^3 + 6A$

3 Use $A^3 = A \cdot A^2 = A \cdot I = A$. ✓ 1
$2A + 6A - 7A = A$

4 Hence option (A).

Answer **(A)** A

Watch out: Putting $A^3 = I$; with $A^2 = I$ we get $A^3 = A$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): For $A = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}$, $A^2 = O$. **Reason (R):** If the product of two matrices is a zero matrix, then at least one of the two matrices must be a zero matrix.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Two non-zero matrices can have a zero product.

1	Find A^2. $A^2 = \begin{bmatrix} 4-4 & -8+8 \\ 2-2 & -4+4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	
2	So Assertion (A) is true.	
3	Here $A \cdot A = O$ but $A \neq O$, so Reason (R) is false.	
4	Hence option (C).	✓ 1

Answer **(C) Assertion (A) is true, but Reason (R) is false.**

Watch out: Carrying the rule "$ab = 0 \Rightarrow a = 0$ or $b = 0$" from numbers over to matrices.

7 **Short answer** Board-pattern 2 marks

Find the values of x and y if $2\begin{bmatrix} x & 3 \\ y & x \end{bmatrix} - \begin{bmatrix} 1 & y \\ x & 4 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & 2 \end{bmatrix}$.

1	Multiply by 2 and subtract. $\begin{bmatrix} 2x-1 & 6-y \\ 2y-x & 2x-4 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & 2 \end{bmatrix}$	✓ ½
2	Equate the $(1,1)$ elements. $2x - 1 = 5 \Rightarrow x = 3$	✓ ½
3	Equate the $(1,2)$ elements. $6 - y = 2 \Rightarrow y = 4$	✓ ½
4	Check the other two elements. $2y - x = 8 - 3 = 5, \quad 2x - 4 = 6 - 4 = 2$	✓ ½

5 Hence $x = 3$ and $y = 4$.

Answer $x = 3, y = 4$

Watch out: Multiplying only the first row by 2 instead of every element.

Level 2

18 marks

8 **Short answer** Board-pattern 2 marks

Construct a 3×3 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \begin{cases} i - j, & \text{if } i \geq j \\ i + j, & \text{if } i < j \end{cases}$

Formula / idea: On and below the diagonal $i \geq j$; above the diagonal $i < j$.

1 Row 1: only a_{11} has $i \geq j$. ✓ ½
$a_{11} = 1 - 1 = 0, \quad a_{12} = 1 + 2 = 3, \quad a_{13} = 1 + 3 = 4$

2 Row 2: a_{21}, a_{22} use $i - j$. ✓ ½
$a_{21} = 2 - 1 = 1, \quad a_{22} = 0, \quad a_{23} = 2 + 3 = 5$

3 Row 3: all use $i - j$. ✓ ½
$a_{31} = 3 - 1 = 2, \quad a_{32} = 3 - 2 = 1, \quad a_{33} = 0$

4 Write the matrix. ✓ ½
$$A = \begin{bmatrix} 0 & 3 & 4 \\ 1 & 0 & 5 \\ 2 & 1 & 0 \end{bmatrix}$$

Answer $A = \begin{bmatrix} 0 & 3 & 4 \\ 1 & 0 & 5 \\ 2 & 1 & 0 \end{bmatrix}$

Watch out: Using $i + j$ on the diagonal; there $i = j$, so the rule $i - j$ applies.

9 **Short answer** PYQ-style 3 marks

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$. Hence find A^{-1}.

1 Find A^2. ✓ 1
$$A^2 = \begin{bmatrix} 9 + 2 & 6 + 2 \\ 3 + 1 & 2 + 1 \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$$

2 Write $A^2 + aA + bI$ and set it equal to O. ✓ ½
$$\begin{bmatrix} 11 + 3a + b & 8 + 2a \\ 4 + a & 3 + a + b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

3 From $4 + a = 0$ and $3 + a + b = 0$. ✓ ½

$a = -4, \quad b = 1$

These also satisfy $11 + 3a + b = 0$ and $8 + 2a = 0$.

4 So $A^2 - 4A + I = O$. Rearrange and take A common. ✓ ½

$A(4I - A) = I \Rightarrow A^{-1} = 4I - A$

Also $(4I - A)A = I$.

5 Compute $4I - A$. ✓ ½

$$A^{-1} = \begin{bmatrix} 4-3 & -2 \\ -1 & 4-1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Answer $a = -4, b = 1$; $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

Watch out: Adding b to every element of $A^2 + aA$; bI adds b only on the diagonal.

10 **Short answer** Board-pattern 3 marks

Express the matrix $A = \begin{bmatrix} 2 & 5 & -3 \\ 1 & 4 & 6 \\ 7 & -2 & 0 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

Formula / idea: $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$: symmetric part plus skew-symmetric part.

1 Write the transpose. ✓ ½

$$A' = \begin{bmatrix} 2 & 1 & 7 \\ 5 & 4 & -2 \\ -3 & 6 & 0 \end{bmatrix}$$

2 Symmetric part $P = \frac{1}{2}(A + A')$. ✓ ½

$$P = \frac{1}{2}\begin{bmatrix} 4 & 6 & 4 \\ 6 & 8 & 4 \\ 4 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 2 \\ 3 & 4 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

3 Check: $P' = P$, so P is symmetric. ✓ ½

4 Skew-symmetric part $Q = \frac{1}{2}(A - A')$. ✓ ½

$$Q = \frac{1}{2}\begin{bmatrix} 0 & 4 & -10 \\ -4 & 0 & 8 \\ 10 & -8 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 4 \\ 5 & -4 & 0 \end{bmatrix}$$

5 Check: $Q' = -Q$, so Q is skew-symmetric. ✓ ½

6 Add to verify: $P + Q = A$.

$$\begin{bmatrix}2 & 3 & 2\\3 & 4 & 2\\2 & 2 & 0\end{bmatrix} + \begin{bmatrix}0 & 2 & -5\\-2 & 0 & 4\\5 & -4 & 0\end{bmatrix} = \begin{bmatrix}2 & 5 & -3\\1 & 4 & 6\\7 & -2 & 0\end{bmatrix} = A$$

✓ ½

Answer $A = \begin{bmatrix}2 & 3 & 2\\3 & 4 & 2\\2 & 2 & 0\end{bmatrix} + \begin{bmatrix}0 & 2 & -5\\-2 & 0 & 4\\5 & -4 & 0\end{bmatrix}$

Watch out: Sign slip in $A - A'$ at the $(1,3)$ place: $-3 - 7 = -10$, not 4.

11 Short answer · Board-pattern · 2 marks

Find the matrices X and Y if $X + 2Y = \begin{bmatrix}4 & 7\\-4 & 5\end{bmatrix}$ and $2X - Y = \begin{bmatrix}3 & -6\\2 & 5\end{bmatrix}$.

Formula / idea: Matrix equations are solved like simultaneous equations: eliminate one matrix.

1 Multiply the second equation by 2 and add the first; Y cancels.

$$5X = \begin{bmatrix}4 & 7\\-4 & 5\end{bmatrix} + \begin{bmatrix}6 & -12\\4 & 10\end{bmatrix} = \begin{bmatrix}10 & -5\\0 & 15\end{bmatrix}$$

✓ ½

2 Divide by 5.

$$X = \begin{bmatrix}2 & -1\\0 & 3\end{bmatrix}$$

✓ ½

3 From the second equation, $Y = 2X - \begin{bmatrix}3 & -6\\2 & 5\end{bmatrix}$.

$$Y = \begin{bmatrix}4 & -2\\0 & 6\end{bmatrix} - \begin{bmatrix}3 & -6\\2 & 5\end{bmatrix}$$

✓ ½

4 Subtract element by element.

$$Y = \begin{bmatrix}1 & 4\\-2 & 1\end{bmatrix}$$

✓ ½

Answer $X = \begin{bmatrix}2 & -1\\0 & 3\end{bmatrix}, Y = \begin{bmatrix}1 & 4\\-2 & 1\end{bmatrix}$

Watch out: Sign slip in $-2 - (-6) = 4$ at the $(1,2)$ place of Y.

12 **Short answer** Board-pattern 3 marks

Find x, if $\begin{bmatrix} x & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ x \\ 2 \end{bmatrix} = O$.

Formula / idea: $(1 \times 3)(3 \times 3) = 1 \times 3$, then $(1 \times 3)(3 \times 1) = 1 \times 1$.

1 Multiply the row matrix by the 3×3 matrix. ✓ 1

$\begin{bmatrix} x+2-1 & 2x+0+0 & 0+1-2 \end{bmatrix} = \begin{bmatrix} x+1 & 2x & -1 \end{bmatrix}$

2 Multiply by the column matrix. ✓ 1

$x(x+1) + x(2x) + 2(-1) = 3x^2 + x - 2$

3 Set the 1×1 matrix equal to O and factorise. ✓ ½

$3x^2 + x - 2 = 0 \Rightarrow (3x-2)(x+1) = 0$

4 Hence the values of x. ✓ ½

$x = \frac{2}{3}$ or $x = -1$

Answer $x = \frac{2}{3}$ **or** $x = -1$

Watch out: Missing that the second entry of the column is also x, which loses the $2x^2$ term.

13 **Short answer** Board-pattern 2 marks

If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, find $\alpha \left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A^2 + I = \sqrt{3}\,A$.

Formula / idea: $\cos^2\alpha - \sin^2\alpha = \cos 2\alpha$, $2\sin\alpha\cos\alpha = \sin 2\alpha$, $1 + \cos 2\alpha = 2\cos^2\alpha$.

1 Multiply A by itself and use the double-angle formulas. ✓ ½

$A^2 = \begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$

2 Add I and write the given equation. ✓ ½

$\begin{bmatrix} 1+\cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & 1+\cos 2\alpha \end{bmatrix} = \begin{bmatrix} \sqrt{3}\cos\alpha & -\sqrt{3}\sin\alpha \\ \sqrt{3}\sin\alpha & \sqrt{3}\cos\alpha \end{bmatrix}$

3 Equate the $(2,1)$ elements; divide by $\sin\alpha \neq 0$. ✓ ½

$2\sin\alpha\cos\alpha = \sqrt{3}\sin\alpha \Rightarrow \cos\alpha = \frac{\sqrt{3}}{2}$

4 In $\left(0, \frac{\pi}{2}\right)$ this gives $\alpha = \frac{\pi}{6}$; the $(1,1)$ element checks. ✓ ½

$$1 + \cos\frac{\pi}{3} = \frac{3}{2} = \sqrt{3}\cos\frac{\pi}{6}$$

Answer $\alpha = \frac{\pi}{6}$

Watch out: Squaring each element of A to get A^2, instead of row-by-column multiplication.

14 **Short answer** Board-pattern 3 marks

If $A' = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.

Formula / idea: $(A')' = A$: transpose A' to get A.

1 Get A by transposing A' (order 3×2). ✓ ½

$$A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \\ 3 & -1 \end{bmatrix}$$

2 Find AB (order 3×2). ✓ 1

$$AB = \begin{bmatrix} 2+0 & -1+0 \\ -4+1 & 2+3 \\ 6-1 & -3-3 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 5 \\ 5 & -6 \end{bmatrix}$$

3 Take the transpose: LHS. ✓ ½

$$(AB)' = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 5 & -6 \end{bmatrix}$$

4 Find $B'A'$: RHS. ✓ ½

$$B'A' = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -4+1 & 6-1 \\ -1 & 2+3 & -3-3 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 5 & -6 \end{bmatrix}$$

5 LHS = RHS, so $(AB)' = B'A'$ is verified. ✓ ½

Answer $(AB)' = B'A' = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 5 & -6 \end{bmatrix}$, **hence verified.**

Watch out: Using the given A' as A in the product AB; it is not even defined that way.

Level 3

23 marks

15 **Long answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $A^2 - 4A - 5I = O$. Hence find A^{-1}.

1 Find A^2. ✓ 1

$$A^2 = \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

2 Substitute in $A^2 - 4A - 5I$. ✓ 1

$$\begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3 Hence $A^2 - 4A - 5I = O$.

4 Rearrange and take A common. ✓ ½

$$A^2 - 4A = 5I \;\Rightarrow\; A(A - 4I) = 5I \;\Rightarrow\; A^{-1} = \frac{1}{5}(A - 4I)$$

$A \cdot \frac{1}{5}(A - 4I) = I.$

5 Compute $A - 4I$ and divide by 5. ✓ ½

$$A^{-1} = \frac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Answer $A^2 - 4A - 5I = O$; $A^{-1} = \frac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$

Watch out: Subtracting 4 from every element of A; $4I$ changes only the diagonal.

16 **Long answer** Board-pattern 3 marks

A stationery shop sells gel pens, ball pens and fountain pens. On Day 1 it sold 20 gel pens, 30 ball pens and 10 fountain pens. On Day 2 it sold 15 gel pens, 40 ball pens and 12 fountain pens. A ball pen costs ₹10 and a fountain pen costs ₹50. The total sales of the two days were ₹2,850. Using matrix multiplication, find the price of a gel pen and the sales on each day.

1 Let a gel pen cost ₹x. Sales matrix (rows: days; columns: gel, ball, fountain). ✓ ½

$$S = \begin{bmatrix} 20 & 30 & 10 \\ 15 & 40 & 12 \end{bmatrix}$$

2 Price column matrix. ✓ ½

$$P = \begin{bmatrix} x \\ 10 \\ 50 \end{bmatrix}$$

3 Sales on each day $= SP$. ✓ 1

$$SP = \begin{bmatrix} 20x + 300 + 500 \\ 15x + 400 + 600 \end{bmatrix} = \begin{bmatrix} 20x + 800 \\ 15x + 1000 \end{bmatrix}$$

4 Total of both days is ₹2,850. ✓ ½

$35x + 1800 = 2850 \Rightarrow x = 30$

5 Put $x = 30$: Day 1 $= 600 + 800 = 1400$; Day 2 $= 450 + 1000 = 1450$. ✓ ½

6 Hence a gel pen costs ₹30; sales were ₹1,400 on Day 1 and ₹1,450 on Day 2.

Answer **Gel pen ₹30; Day 1 sales ₹1,400; Day 2 sales ₹1,450**

Watch out: Writing prices as a row matrix, so that SP is not defined.

17 **Long answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, find $A^2 - 5A - 14I$. Hence obtain A^3.

1 Find A^2. ✓ 1

$$A^2 = \begin{bmatrix} 9 + 20 & -15 - 10 \\ -12 - 8 & 20 + 4 \end{bmatrix} = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix}$$

2 Substitute in $A^2 - 5A - 14I$. ✓ ½

$$\begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

3 So $A^2 = 5A + 14I$. Multiply both sides by A. ✓ ½

$A^3 = 5A^2 + 14A$

4 Substitute A^2 and A. ✓ 1

$$A^3 = \begin{bmatrix} 145 & -125 \\ -100 & 120 \end{bmatrix} + \begin{bmatrix} 42 & -70 \\ -56 & 28 \end{bmatrix} = \begin{bmatrix} 187 & -195 \\ -156 & 148 \end{bmatrix}$$

Same as $39A + 70I$, using $A^2 = 5A + 14I$ again.

Answer $A^2 - 5A - 14I = O$; $A^3 = \begin{bmatrix} 187 & -195 \\ -156 & 148 \end{bmatrix}$

Watch out: Writing $A^3 = 5A^2 + 14$; multiplying $14I$ by A gives $14A$, not 14.

18 **Long answer** PYQ-style 5 marks

If $A = \begin{bmatrix} \cos^2\alpha & \cos\alpha\sin\alpha \\ \cos\alpha\sin\alpha & \sin^2\alpha \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2\beta & \cos\beta\sin\beta \\ \cos\beta\sin\beta & \sin^2\beta \end{bmatrix}$, show that AB is the zero matrix when α and β differ by an odd multiple of $\dfrac{\pi}{2}$.

Formula / idea: $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha-\beta)$.

1	Element $(1,1)$ of AB; take $\cos\alpha\cos\beta$ common. $\cos^2\alpha\cos^2\beta + \cos\alpha\sin\alpha\cos\beta\sin\beta$ $= \cos\alpha\cos\beta\,(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \cos\alpha\cos\beta\cos(\alpha-\beta)$	✓ 1
2	Element $(1,2)$; take $\cos\alpha\sin\beta$ common. $\cos^2\alpha\cos\beta\sin\beta + \cos\alpha\sin\alpha\sin^2\beta$ $= \cos\alpha\sin\beta\,(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \cos\alpha\sin\beta\cos(\alpha-\beta)$	✓ 1
3	Element $(2,1)$; take $\sin\alpha\cos\beta$ common. $\cos\alpha\sin\alpha\cos^2\beta + \sin^2\alpha\cos\beta\sin\beta = \sin\alpha\cos\beta\cos(\alpha-\beta)$	✓ ½
4	Element $(2,2)$; take $\sin\alpha\sin\beta$ common. $\cos\alpha\sin\alpha\cos\beta\sin\beta + \sin^2\alpha\sin^2\beta = \sin\alpha\sin\beta\cos(\alpha-\beta)$	✓ ½
5	Take $\cos(\alpha-\beta)$ out of the matrix. $AB = \cos(\alpha-\beta)\begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta \end{bmatrix}$	✓ ½
6	α and β differ by an odd multiple of $\frac{\pi}{2}$. $\alpha-\beta = (2n+1)\dfrac{\pi}{2},\ n\in\mathbb{Z} \Rightarrow \cos(\alpha-\beta) = 0$	✓ 1
7	Every element of AB is then 0. $AB = 0\cdot\begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta \end{bmatrix} = O$	✓ ½
8	Hence AB is the zero matrix.	

Answer $AB = \cos(\alpha-\beta)\begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta \end{bmatrix} = O$ **when** $\alpha-\beta = (2n+1)\dfrac{\pi}{2}$

Watch out: Using $\cos(\alpha+\beta)$; the bracket is $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha-\beta)$.

19 **Long answer** PYQ-style 5 marks

If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$, show that $A^3 = pI + qA + rA^2$.

1 Find $A^2 = A \cdot A$. ✓ 1½

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix}$$

Row 3: $(0+0+pr,\ p+0+qr,\ 0+q+r^2)$.

2 Find $A^3 = A \cdot A^2$. ✓ 1½

$$A^3 = \begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+pr^2 & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$$

Row 3 is $p(\text{row } 1) + q(\text{row } 2) + r(\text{row } 3)$ of A^2.

3 Write pI, qA and rA^2. ✓ ½

$$pI = \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix}, \quad qA = \begin{bmatrix} 0 & q & 0 \\ 0 & 0 & q \\ pq & q^2 & qr \end{bmatrix}, \quad rA^2 = \begin{bmatrix} 0 & 0 & r \\ pr & qr & r^2 \\ pr^2 & pr+qr^2 & qr+r^3 \end{bmatrix}$$

4 Add them: RHS. ✓ 1

$$pI + qA + rA^2 = \begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+pr^2 & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$$

5 LHS = RHS element by element. ✓ ½

$$A^3 = pI + qA + rA^2$$

Answer **Both sides equal** $\begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+pr^2 & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$**, so** $A^3 = pI + qA + rA^2$**.**

Watch out: Adding p to every element for pI; it adds p only on the diagonal.

20 Case study Board-pattern 4 marks

Case Study: A dairy cooperative sells milk, curd and paneer in two towns P and Q. In one week, town P buys 500 litres of milk, 150 kg of curd and 40 kg of paneer; town Q buys 400 litres of milk, 200 kg of curd and 50 kg of paneer. The selling prices are ₹60 per litre of milk, ₹80 per kg of curd and ₹400 per kg of paneer. The cost prices are ₹45, ₹60 and ₹320 respectively.

(i) Write the weekly sales as a 2×3 matrix and the selling prices as a column matrix. 1 mark

(ii) Find the revenue from town P. 1 mark

(iii)(a) Using matrix multiplication, find the profit from each town and the total profit. 2 marks

OR

(iii)(b) The cooperative changes the price of paneer to ₹x per kg, keeping the other prices the same. The total revenue from both towns becomes ₹1,22,500. Using matrices, find x. 2 marks

(i) 1 mark

1 Rows = towns P, Q; columns = milk, curd, paneer.

$$S = \begin{bmatrix} 500 & 150 & 40 \\ 400 & 200 & 50 \end{bmatrix}$$

✓ ½

2 Selling prices as a column matrix.

$$P = \begin{bmatrix} 60 \\ 80 \\ 400 \end{bmatrix}$$

✓ ½

Answer (i) $S = \begin{bmatrix} 500 & 150 & 40 \\ 400 & 200 & 50 \end{bmatrix}$, $P = \begin{bmatrix} 60 \\ 80 \\ 400 \end{bmatrix}$

(ii) 1 mark

1 Multiply town P's row by the price column.

$$\begin{bmatrix} 500 & 150 & 40 \end{bmatrix} \begin{bmatrix} 60 \\ 80 \\ 400 \end{bmatrix} = 30000 + 12000 + 16000$$

✓ ½

2 Revenue from town P is ₹58,000.

$$= 58000$$

✓ ½

Answer (ii) **₹58,000**

(iii)(a) 2 marks

1 Profit per unit = selling price − cost price.

$$\begin{bmatrix} 60 \\ 80 \\ 400 \end{bmatrix} - \begin{bmatrix} 45 \\ 60 \\ 320 \end{bmatrix} = \begin{bmatrix} 15 \\ 20 \\ 80 \end{bmatrix}$$

✓ ½

2 Profit matrix = S times the profit column. ✓ 1

$$\begin{bmatrix}500 & 150 & 40\\ 400 & 200 & 50\end{bmatrix}\begin{bmatrix}15\\20\\80\end{bmatrix}=\begin{bmatrix}7500+3000+3200\\6000+4000+4000\end{bmatrix}=\begin{bmatrix}13700\\14000\end{bmatrix}$$

3 Total profit $= 13700 + 14000 = 27700$. ✓ ½

Answer (iii)(a) **Town P: ₹13,700; town Q: ₹14,000; total ₹27,700**

OR

(iii)(b)

2 marks

1 New price column has x for paneer. ✓ 1

$$\begin{bmatrix}500 & 150 & 40\\ 400 & 200 & 50\end{bmatrix}\begin{bmatrix}60\\80\\x\end{bmatrix}=\begin{bmatrix}42000+40x\\40000+50x\end{bmatrix}$$

2 Total revenue is ₹1,22,500. ✓ ½

$82000 + 90x = 122500 \Rightarrow 90x = 40500$

3 Hence the new price of paneer. ✓ ½

$x = 450$

Answer (iii)(b) $x = 450$**, i.e. ₹450 per kg**

Watch out: Using the selling prices instead of (selling − cost) when finding profit.